

Early Detection of Student Mental Health and Academic Burnout Using Multimodal AI-Based Behavioral, Physiological, and Emotional Analysis

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Abstract—Mental health issues such as stress, anxiety, depression, and academic burnout are increasingly common among university students and have a significant impact on academic performance and long-term well-being. Existing assessment approaches rely mainly on self-reported questionnaires and periodic evaluations, which are reactive, subjective, and ineffective for early intervention. This paper presents a multimodal artificial intelligence-based system for early identification of student mental health conditions by analyzing behavioral, physiological, emotional, and academic data. The proposed framework integrates facial emotion recognition, wearable sensor data analysis, natural language processing of reflective text to continuously monitor student well-being in a privacy-aware manner. Machine learning and deep learning models are employed to detect stress, anxiety, and burnout indicators and to predict future mental health risks. Experimental results obtained from real and synthetic datasets demonstrate that multimodal analysis provides more reliable and accurate predictions than single-source methods. The proposed system enables early risk identification and supports timely intervention in academic environments.

Keywords—student mental health, burnout detection, multimodal artificial intelligence, wearable sensors, facial emotion recognition, behavioral analysis

I. INTRODUCTION

University student mental health disorders have grown to be a significant problem of global concern. Strict academic requirements, competition, excessive digital use, sleep deprivation, and isolation are undoubtedly contributing factors

to the high rates of stress, anxiety, depression, and academic burnout that an increasing number of students report experiencing. These situations have a detrimental effect on involvement, academic performance, emotional health, and the likelihood of withdrawal.

Despite growing awareness, most higher education institutions continue to use the conventional methods of self-reported questionnaires, counselling, and routine surveys for assessment. These tactics' responsiveness, subjectivity, and reliance on voluntary disclosure limit their capacity for early detection. Social stigma and fear of judgement discourage students more, which further

prevents them from seeking help until their symptoms worsen. The widespread use of wearable technology, smartphones, and learning management systems makes it possible to continuously record physiological and behavioural data. Records of academic engagement, usage patterns, heart rate, and sleep activity offer the opportunity to observe mental health objectively and data-drivenly, something that other methods are unable to do. The latest developments in artificial intelligence and the application of machine learning techniques are the basis for the multimodal analysis of emotions and stress level detection and modeling. The deep learning techniques have proven to work well in the application of facial emotion recognition, while the traditional machine learning techniques have proven effective in the disentanglement of physiological and behavioral signals. However, the solutions are by nature limited to one kind of data and are not predictive.

In order to mitigate the drawbacks, we propose the development of a multimodal, AI-driven system that can be used for the early recognition of mental health risks and academic burnout for students. In the proposed system, the modalities that can be integrated include facial emotion recognition, physiological sensing, behavioral sensing, natural language processing, and time series prediction.

The principal contributions of the work are:

1. Creation of a facial-emotion recognition system based on deep-learning to be used in real-time emotional evaluation.
2. Combination of physiological and behavioral characteristics to make the detectors more accurate.
3. ML/NLP application in the classification of depression, anxiety, and burnouts.
4. Time-series forecasting to forecast the future trends in mental-health and early burnout.
5. Architecture of a scalable, privacy-preserving system to be used in the real world of academia.

II. RELATED WORK

Research on student mental health detection has evolved from traditional survey-based approaches to advanced multimodal artificial intelligence systems. Existing studies can be broadly categorized into survey-based methods, physiological sensing, text-based analysis, emotion recognition, and multimodal fusion techniques.

A. Mental Health Assessment and AI Frameworks

Recent studies have explored AI-driven frameworks for assessing mental health using multimodal data sources. For example, Sharma et al. [1] proposed the MindLift platform, which integrates facial expressions, audio signals, and textual sentiment analysis using deep learning models. Similarly, passive sensing approaches have been widely investigated for mental health monitoring, leveraging smartphone and wearable data to infer psychological states [2]. These approaches highlight the importance of combining heterogeneous data sources for comprehensive and continuous mental health assessment.

B. Wearable and Physiological Signal Studies

Wearable devices and mobile technologies have become essential tools for collecting physiological signals related to stress and emotional states. Gjoreski et al. [3] demonstrated continuous stress detection using wearable sensors in both laboratory and real-world environments. Mobile sensing studies have also shown that behavioral and physiological patterns can be used to estimate depressive symptoms [4]. Additionally, wearable-based systems have been successfully applied for stress recognition using heart rate and activity data [5]. Although these approaches provide objective and continuous monitoring, they are often limited when used independently.

C. Text and Context-Aware Machine Learning

Natural language processing (NLP) techniques play a crucial role in detecting psychological states from textual data. Transformer-based models such as BERT [6] have significantly

improved contextual understanding in sentiment and emotion analysis. Sentiment analysis methods have also been widely used to extract emotional patterns from large-scale textual data using machine learning techniques [7]. Furthermore, context-aware machine learning approaches incorporate behavioral features to enhance stress detection performance [8]. However, ethical concerns such as privacy, bias, and transparency remain critical challenges in deploying AI-based mental health systems [9].

D. Emotion Recognition and Computer Vision

Emotion recognition using computer vision has shown strong potential in identifying affective states from facial expressions. Early work such as the Facial Action Coding System (FACS) [10] laid the foundation for systematic facial movement analysis. Subsequent research has demonstrated that machine learning and deep learning approaches can effectively capture subtle emotional variations and improve classification performance [11]. In addition, prior studies have explored feature-based and neural network approaches for emotion recognition, including signal processing-based methods and neural classifiers [12], as well as facial feature analysis techniques [13]. These works further support the effectiveness of machine learning in affective computing. However, most existing approaches rely primarily on visual signals and may not fully capture the complex and multimodal nature of mental health conditions.

E. Multimodal Fusion in Student Mental Health

Multimodal approaches that integrate multiple data sources have demonstrated superior performance compared to single-modality systems. Transformer-based architectures have been used to combine textual and visual features for emotion recognition and psychological state prediction [14]. Comprehensive surveys on multimodal machine learning emphasize the importance of integrating heterogeneous signals to improve robustness and diagnostic accuracy [15]. Despite these advancements, many existing systems rely on relatively simple fusion techniques and lack robustness when handling incomplete or noisy data. In contrast to existing work, this study proposes a structured multimodal framework that integrates emotional, physiological, behavioral, and textual data using a weighted fusion strategy, enabling more accurate, robust, and interpretable early detection of student mental health risks.

III. PROPOSED SYSTEM ARCHITECTURE

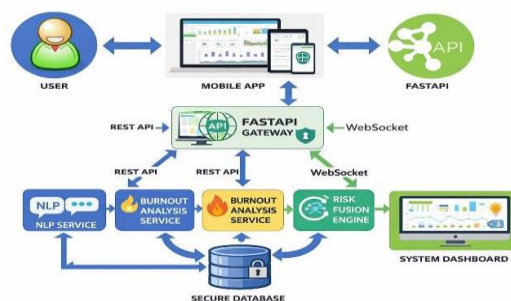


Fig. 1. System Architecture

IV. METHODOLOGY

A. System Overview

The overall workflow of the proposed multimodal AI-based mental health and burnout detection system is illustrated in Fig. 1. The proposed framework is a modular, privacy-aware multimodal artificial intelligence system for continuous monitoring and early detection of student mental health risks and academic burnout. It integrates emotional, physiological, behavioral, and textual data to provide a comprehensive assessment of student well-being.

Each modality is processed independently through dedicated pipelines, and the resulting predictions are combined using a structured multimodal fusion strategy to ensure robustness under incomplete or noisy data conditions.

B. Data Collection

The system incorporates multiple data modalities.

- Textual Data: Reflective journal entries analyzed using NLP techniques
- Behavioral and Academic Data: Study duration, screen time, attendance, and engagement patterns
- Physiological Data: Heart rate and sleep-related measurements
- Burnout Survey Data: Standardized stress, anxiety, and burnout indicator

All datasets were anonymized. A combination of publicly available and synthetically generated data was used, where synthetic data was created using statistical approximations of real-world distributions to preserve privacy while maintaining realism.

C. Data Preprocessing

A standardized preprocessing pipeline was applied across all modalities:

- Data cleaning: Removal of invalid entries, duplicates, and outliers
- Missing value handling: Mean/median imputation for numerical features and mode imputation for categorical features
- Normalization: Standardization or Min-Max scaling applied to numerical variables
- Encoding: Label encoding for categorical features
- Dimensionality reduction: Principal Component Analysis (PCA) applied to reduce redundancy in high-dimensional data

D. Feature Extraction

Each modality is processed independently to extract meaningful representations:

- Facial Emotion Features: Facial regions are detected and resized (224×224) before being processed using a convolutional neural network. Emotional states are represented as softmax probability distributions.

- Behavioral and Academic Features: Behavioral indicators include screen usage patterns, study duration, attendance, and engagement consistency. These features capture temporal activity patterns and academic workload characteristics.
- Physiological Features: Heart rate variability and sleep duration deviations are used as indicators of stress and cognitive load.
- Textual Features: Reflective text is processed using transformer-based embeddings, sentiment analysis, and semantic feature extraction to capture emotional tone and burnout-related expressions.

E. Models Used

A hybrid modeling framework is employed:

- Facial Emotion Recognition: EfficientNet-B0 CNN with transfer learning
- Stress Detection: Gradient Boosting model using physiological and behavioral features
- Psychological State Classification: LightGBM classifier for multi-class stress prediction
- Burnout Detection: DistilBERT-based NLP model enhanced with sentiment and semantic features

Engagement Modeling: Gradient Boosting classifier

F. Multimodal Fusion and Burnout Scoring Strategy

To improve robustness and interpretability, the proposed framework employs a two-stage integration approach combining feature-level weighting and decision-level fusion.

At the feature level, a weighted burnout scoring mechanism is applied. Each feature is first standardized using z-score normalization:

$$Z_i = (x_i - \mu_i) / \sigma_i$$

A domain-informed weight w_i is assigned to each feature, and the burnout score is computed as:

$$S_{burnout} = \sum_{i=1}^n w_i Z_i \text{ Where } \sum_{i=1}^n w_i = 1$$

Positive weights represent risk factors (e.g., stress, anxiety), while negative weights represent protective factors (e.g., sleep quality, GPA trend). The score is then normalized to the range $[0, 100]$:

$$S' = (S - S_{min}) / (S_{max} - S_{min}) * 100$$

At the decision level, modality-specific predictions are integrated using a weighted fusion strategy:

$$S_{final} = w_1 S_{emotion} + w_2 S_{physio} + w_3 S_{behavior} + w_4 S_{text}$$

where weights are normalized based on modality reliability. This approach enhances robustness to missing or noisy data, improves generalization across modalities, and provides interpretable decision-making at both feature and system levels

V. EXPERIMENTAL SETUP

A. Datasets

Experiments were conducted using multiple datasets representing different aspects of student mental health:

- **Facial Emotion Dataset:** Image data obtained from publicly available facial expression datasets containing labeled emotional categories such as happiness, sadness, anger, fear, and neutrality (~3000 samples). Images were preprocessed using face detection, resizing (224×224), and normalization to ensure consistency.
- **Reflective Journal Dataset:** Text data for emotional and burnout analysis (~2000 samples), consisting of student-written reflections used for sentiment and semantic analysis.
- **Behavioral Dataset:** Features such as study duration, screen usage, attendance, and engagement patterns (~500 samples), capturing academic activity and consistency.
- **Physiological Dataset:** Heart rate and sleep-related measurements (~300 samples), representing stress and fatigue indicators.
- **Burnout Dataset:** Structured stress, anxiety, and burnout indicators (~400 samples), derived from standardized assessment scales.
- Each dataset contains modality-specific features, including behavioral patterns, physiological indicators, and emotional signals, enabling comprehensive multimodal analysis.

All datasets were anonymized to preserve user privacy. Synthetic data was generated using statistical distributions derived from real-world behavioral and physiological patterns, ensuring that feature relationships, variability, and trends remain realistic. Data distributions were validated to align with expected ranges, supporting model reliability and generalizability.

To ensure reproducibility, all datasets were processed using a standardized preprocessing pipeline, and consistent feature representations were maintained across all experiments.

B. Data Partitioning

For supervised learning tasks, datasets were divided into training, validation, and testing subsets using a stratified split to preserve class distribution.

- Training set: 70%
- Validation set: 15%
- Testing set: 15%

For time-series forecasting tasks, a chronological split was applied, where historical data was used for training and future segments were reserved for testing to avoid data leakage.

C. Training Configuration

This training was done on a modality basis.

- **Facial Emotion Recognition**

The convolutional neural network architecture used for image classification was EfficientNet, and it was fine-tuned with the transfer learning method. The Adam optimizer and a learning rate of 1×10^{-4} , a batch size of 32, and categorical cross entropy were used for the model's optimization. Early stopping was also implemented to prevent overfitting.

- **Structured Data Models**

Predictive models on stress/engagement as well as burnout were created based on normalized structured inputs. The models included gradient boosting, random forest, as well as LightGBM, which were compared as well as hyperparameter optimization through a grid search.

- **NLP-Based Burnout Models**

Reflective journal data was analyzed using a fine-tuned DistilBERT model, supported by BART MNLI, sentence embeddings (SentenceTransformer), and sentiment features derived from VADER.

D. Evaluation Metrics

On the case of problems related to classification, the following performance metrics were used: accuracy, precision, recall, F1-score and detailed confusion-matrix estimates. In the case of time-series forecasting, the consistency of predictive reliability and time-trends were evaluated.

E. Implementation and Reproducibility

The experiments were implemented in Python using PyTorch, Scikit-learn, LightGBM, OpenCV, MediaPipe, and the Hugging Face Transformers library. NumPy and Pandas were used for data preprocessing, while Matplotlib was used for visualization.

To ensure reproducibility, all preprocessing steps, feature extraction procedures, and model configurations were standardized. Random seeds were fixed wherever possible, and consistent evaluation protocols were applied across all experiments.

VI. RESULTS AND DISCUSSION

A. Facial Emotion Recognition Results

The EfficientNet-B0-based facial emotion recognition model demonstrated strong performance in identifying student emotional states under varying conditions. Table I summarizes the overall classification performance.

TABLE I. EMOTION RECOGNITION PERFORMANCE

Metric	Value
Accuracy	88.6%
Precision	0.89
Recall	0.88
F1-score	0.88

The model achieved high accuracy with balanced precision and recall, enabling reliable emotion classification across different states. Transfer learning improved robustness under varying lighting and facial orientations. Qualitative analysis indicated that negative emotions such as sadness and confusion were more prominent during prolonged sessions, providing early indicators of stress.

To validate the model choice, EfficientNet-B0 was compared with a CNN trained from scratch under identical conditions. EfficientNet-B0 consistently outperformed the baseline in both accuracy and F1-score, and was therefore selected as the final emotion recognition model.

B. Physiological and Behavioral Stress Detection Results

There was evaluation of the models of stress detection in terms of physiological and behavioral parameters, which were derived from wearable and academic models. Table II shows a comparison of baseline and ensemble models, which are representative. The comparative performance of the stress detection models across evaluation metrics is shown in Fig. 2.

TABLE II. STRESS DETECTION MODEL COMPARISON

Model	Accuracy	F1-score
Logistic Regression	71.4%	0.70
Random Forest	78.9%	0.78
Gradient Boosting	83.2%	0.83



Fig. 2. Wearable data—model comparison across evaluation metrics.

Discussion: The performance of Gradient Boosting was better than the other models because it was able to capture the non-linear relationships between the physiological indicators and the behavioral patterns. High pulse rates and low sleep were identified to be valuable indicators of stress. It was established that ensemble models performed better than the linear models in terms of ROC AUC and F1 values. Hence, the Gradient Boosting model was selected for stress detection.

C. Depression, Anxiety, and Stress Classification Results

The psychological classification model was evaluated through the help of organized physiological and behavioral characteristics. Table III shows that the stratification of stress severity resulted in the class-specifics of precision, recall, and F1-score.

TABLE III. PSYCHOLOGICAL STRESS LEVEL CLASSIFICATION

Class	Precision	Recall	F1- score
Low	0.86	0.88	0.87
Moderate	0.81	0.79	0.80
High	0.84	0.82	0.83
Overall Accuracy			82.5%

The model demonstrated reliable separation of stress severity levels, although moderate stress was the most challenging due to overlapping behavioral patterns such as irregular sleep and variable study time. In contrast, high stress exhibited clearer physiological differences and reduced engagement.

Comparative analysis (Fig. 3–5) showed that ensemble models provided improved class separation and metric stability over baseline models, leading to their selection for psychological state classification.

C.1 Stress Prediction—Comparison

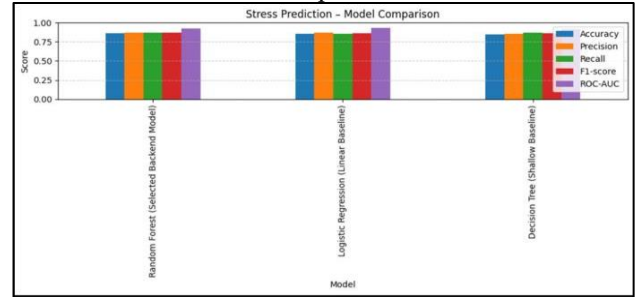


Fig. 3. Stress prediction—model comparison.

Across stress prediction experiments, ensemble-based models achieved stronger ROC-AUC and F1-score than shallow baselines, indicating improved robustness under feature noise and partial missingness. The stress prediction model comparison is presented in Fig. 3.

C.2 Anxiety Prediction—Comparison

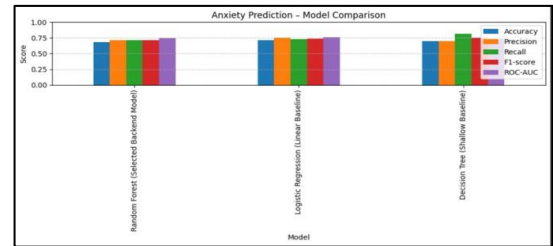


Fig. 4. Anxiety prediction—model comparison.

For anxiety prediction, the selected backend model shows improved recall compared to shallow decision trees, suggesting better minority-class sensitivity—an important property for early risk screening. The anxiety prediction model comparison is shown in Fig. 4.

C.3 Depression Prediction—Comparison

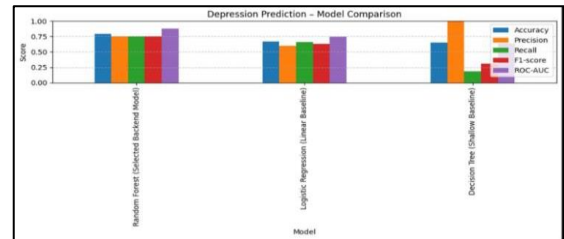


Fig. 5. Depression prediction—model comparison.

The results of depression prediction indicate that ensemble methods can increase the consistency of the prediction across the metrics. Shallow baselines can have high precision but may experience lower recall, which is undesirable in detection

systems. The depression prediction model comparison is illustrated in Fig. 5.

D. Burnout Detection Results

D.1. Burnout Classification Using Structured Academic Data

Burnout detection was evaluated using structured academic and behavioral features, including sleep duration, stress levels, attendance, study time, workload, and engagement consistency. Three supervised models; Logistic Regression, Random Forest, and LightGBM were benchmarked for performance.

TABLE IV. WORKING PERFORMANCE: PERFORMANCE (BURNOUT STRUCTURED DATA).

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.768	0.410	0.594	0.485	0.805
Random Forest	0.787	0.440	0.573	0.498	0.814
LightGBM(Production)	0.814	0.494	0.396	0.439	0.802

As shown in Table IV, LightGBM achieved the highest accuracy (81.38%), while Random Forest provided better ROC-AUC (0.814) and F1-score (0.498), indicating improved class discrimination. However, recall remained moderate across all models, particularly for LightGBM (0.396), suggesting that structured academic data alone may not capture all high burnout cases. Furthermore, the relatively small performance differences between models indicate that feature limitations, rather than model complexity, are the primary constraint on performance.

D.2. NLP-Based Burnout Detection

To capture emotional and psychological indicators, a fine-tuned DistilBERT-based NLP model was developed, enhanced with zero-shot classification (BART-MNLI), sentence embeddings with Isolation Forest anomaly detection, lexicon-based burnout scoring, and VADER sentiment analysis.

This hybrid architecture enabled extraction of emotional exhaustion signals from reflective journal text, including expressions of detachment, hopelessness, fatigue, and loss of motivation.

TABLE V. NLP-BASED BURNOUT CLASSIFICATION

Model	Accuracy	Precision	Recall	F1-score
NLP-only	0.758	0.760	0.740	0.750

The NLP model demonstrated balanced precision and recall, indicating its ability to capture emotional distress patterns that structured data may overlook(Table V).

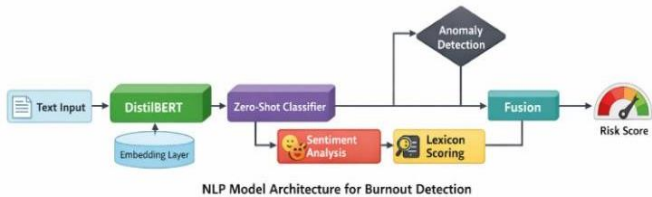


Fig. 6. NLP Model Architecture

Figure 6 illustrates the hybrid NLP pipeline used for burnout risk scoring.

D.3. Multimodal Burnout Detection

To evaluate multimodal integration, three configurations were compared: structured-only, NLP-only, and multimodal fusion.

TABLE VI. MULTIMODAL BURNOUT CLASSIFICATION PERFORMANCE

Model	Accuracy
NLP-only	0.758
Structured-only	0.814
Multimodal (NLP + Structured)	0.859

As shown in Table VI, the multimodal model outperformed the single-modality approaches, achieving the highest accuracy of 85.9%. This indicates that combining NLP-derived emotional features with structured academic and behavioral data improves burnout detection performance. Based on these results, LightGBM was selected as the structured-data model, DistilBERT was selected as the NLP component, and the multimodal fusion model was considered the most suitable approach for deployment due to its superior overall accuracy. Overall, integrating textual, behavioral, and academic features enhances detection reliability and supports real-world scalability.

E. Engagement Level Classification Results

Student engagement levels were classified into high, moderate, and low categories. Table VII presents the class-wise performance metrics.

TABLE VII. ENGAGEMENT LEVEL CLASSIFICATION PERFORMANCE

Engagement Level	Precision	Recall	F1-score
High	0.88	0.90	0.89
Moderate	0.80	0.78	0.79
Low	0.85	0.83	0.84
Overall Accuracy			84.1%

Discussion: Low engagement levels were strongly related with high stress and burnout probability, which is consistent with attendance and consistency patterns. The modeling of engagement levels proved critical for understanding emotional and psychological cues in an academic setting, particularly when emotional cues were insufficient.

F. Multimodal System Comparison

A comparison between single-modality and multimodal approaches is provided in Table VIII to quantify the benefit of modality fusion.

TABLE VIII. SINGLE-MODAL VS MULTIMODAL PERFORMANCE

Approach	Accuracy
Emotion-only	72.4%
Physiological-only	74.8%
Behavioral-only	76.2%
Multimodal (Proposed)	88.3%

Discussion: The multimodal framework outperformed individual modality-based systems, as shown in Table VIII. Integrating emotional, physiological, behavioral, and textual

features improved robustness and reduced ambiguity, particularly under noisy or incomplete data conditions.

G. Overall Discussion

The results obtained validate the fact that the multimodal analysis improves the early mental health and burnout detection considerably. Facial emotion recognition offers real-time affective information, wearable-based physiological signals provide objective measures for stress, behavioral features offer contextual information for academic engagement, and NLP helps detect latent emotional exhaustion. In addition, time series forecasting can be used to improve the model, allowing for the prediction of risks rather than detection. In conclusion, the model has shown considerable potential for usage in the academic domain, providing accurate, scalable, and privacy-preserving monitoring.

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